

**School of Electrical Engineering and Computer Science**  
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## RESEARCH SUMMARY

I am a Ph.D. candidate in Computer Science developing data-efficient machine learning methods for scientific discovery and decision-making under scarce, offline, and expensive data. My research spans probabilistic modeling, Bayesian and offline optimization, generative optimization, and adaptive experimental design, with applications in materials discovery, advanced manufacturing, and wearable systems. My work has been published at ICML, AISTATS, AAAI, and IAAI.

## EDUCATION

**Washington State University**, Pullman, WA 2022 – 2027 (Expected)  
Doctor of Philosophy in Computer Science (GPA: 3.96/4.0)  
Advisors: Prof. Jana Doppa and Prof. Nghia Hoang  
Dissertation: *Use-inspired AI for Engineering Design and Scientific Discovery*

**Washington State University**, Pullman, WA 2022 – 2025  
Master of Science in Computer Science (GPA: 3.97/4.0)  
Advisors: Prof. Jana Doppa and Prof. Nghia Hoang  
Thesis: *Nanoporous Materials Discovery via Search Bias-Guided Surrogate Modeling*

**École Polytechnique de Tunisie**, Tunis, Tunisia 2019 – 2022  
Engineering Degree in Computer Science, Major: Signals and Systems  
*Graduated with Excellence*

**Preparatory School for Engineering Studies of Tunis (IPEIT)**, Tunisia 2017 – 2019  
University First Cycle Studies, Dual Degree: Mathematics-Physics  
*Graduated with 76th National Rank*

## AWARDS AND HONORS

**Founders' Award for Outstanding Masters Thesis Student**, Association for Faculty Women 2026

**Innovative Deployed AI Application Award**, Association for the Advancement of Artificial Intelligence (AAAI) / IAAI 2026

**AAAI Scholarship and Volunteer Award**, Association for the Advancement of Artificial Intelligence (AAAI) 2026

**WSU Graduate & Professional Student Travel Grant (AAAI 2026)**, Graduate & Professional Student Association (GPSA), Washington State University 2026

**Outstanding Graduate Teaching Assistant Award**, Voiland College of Engineering,  
Washington State University 2025

**Outstanding Graduate Teaching Assistant Award**, EECS Department, Washington  
State University 2025

**ICML Scholarship Award**, International Conference on Machine Learning (ICML), Vienna,  
Austria 2024

**WSU Graduate & Professional Student Travel Grant (ICML 2024)**, Graduate &  
Professional Student Association (GPSA), Washington State University 2024

**Nakhara Scholarship**, Voiland College of Engineering and Architecture, Washington State  
University 2022 – 2023

**Tunisian Government Scholarship for Engineering Study**, awarded for ranking in the  
*Top 5% nationally* in the Qualification Exam for Engineering Schools Entrance 2019 – 2022

## PUBLICATIONS

### PUBLISHED PAPERS

1. Minh Hoang, **Azza Fadhel**, Aryan Deshwal, Janardhan Rao Doppa, Trong Nghia Hoang. **Learning Surrogates for Offline Black-Box Optimization via Gradient Matching**. *Proceedings of International Conference on Machine Learning (ICML)*, 2024. (27% acceptance rate)
2. **Azza Fadhel**, Nathaniel W. Zuckschwerdt, Aryan Deshwal, Susmita Bose, Amit Bandyopadhyay, Jana Doppa. **Discovery of Feasible 3D Printing Configurations for Metal Alloys via AI-Driven Adaptive Experimental Design**. *Proceedings of AAAI Conference on Artificial Intelligence*, 2026. (24% acceptance rate)
3. **Azza Fadhel**, Yassine Chemingui, Minh Hoang, Aryan Deshwal, Trong Nghia Hoang, Jana Doppa. **Nanoporous Materials Discovery via Search Bias-Guided Surrogate Modeling**. *Proceedings of AAAI Conference on Artificial Intelligence (AAAI)*, 2026. (24% acceptance rate)
4. **Azza Fadhel**, The Hung Tran, Trong Nghia Hoang, Janardhan Rao Doppa. **Black-Box Optimization from Small Offline Datasets via Meta Learning with Synthetic Tasks**. *Proceedings of International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026. (25% acceptance rate)
5. Nathaniel W. Zuckschwerdt, **Azza Fadhel**, Aryan Deshwal, Janardhan Rao Doppa, Susmita Bose, Amit Bandyopadhyay. **Artificial Intelligence Driven Approach to Accelerate Discovery of Directed Energy Deposition of GRCop-42**. *Journal of Virtual and Physical Prototyping*, 2026.
6. **Azza Fadhel**, Nathaniel W. Zuckschwerdt, Aryan Deshwal, Susmita Bose, Amit Bandyopadhyay, and Janardhan Rao Doppa. **AI-Driven Discovery of Feasible 3D Printing Configurations for Metal Alloys**. *AI Magazine*, 47(3), e70069, 2026. *Invited Special Topic Article*.

### PAPERS UNDER REVIEW

1. The Hung Tran, **Azza Fadhel**, Quang Hung Pham, Manh Cuong Dao, Phi Le Nguyen, Jana Doppa, Trong Nghia Hoang. **Boosting Offline Reinforcement Learning in Data-Scarce**

**Settings via Guided Diffusion Distillation.** *Proceedings of AAAI Conference on Artificial Intelligence (AAAI)*, 2027.

2. **Azza Fadhel\***, Fatemeh Alizadehziri\*, Ganapati Bhat, and Janardhan Rao Doppa. **OptWear: AI-Driven Human-Centered Design of Wearable Health Systems.** *Proceedings of the AAAI Conference on Artificial Intelligence*, 2027. \*Equal contribution.
3. **Azza Fadhel\***, Fatemeh Alizadehziri\*, Ganapati Bhat, and Janardhan Rao Doppa. **Wear-DSE: Cost-Aware Multi-Fidelity Design Space Exploration for Wearable Health Systems.** *ACM Transactions on Embedded Computing Systems (TECS)*, 2026. \*Equal contribution.

## MANUSCRIPTS IN PREPARATION

1. **Azza Fadhel**, Trong Nghia Hoang, and Janardhan Rao Doppa. **Generative Optimization for Offline Rare-Positive Scientific Discovery.** Manuscript in preparation, 2026.
2. **Azza Fadhel**, Qia Ke, Qiong Zhou, Cory Simon, Eric Douglas Bloch, Emma Scurek, and Janardhan Rao Doppa. **Data-Efficient Discovery of PFAS-Capturing Materials via Constrained Multi-Objective Bayesian Optimization.** Manuscript in preparation, 2026.

## RESEARCH EXPERIENCE

**Graduate Research Assistant**, Washington State University, EECS      September 2022 – Present

- Developed machine learning-based surrogate models from experimental data for offline black-box optimization using gradient-matching.
- Created search bias-guided surrogate modeling frameworks combining value-matching and optimization-regularized objectives to accelerate discovery of high-performing materials.
- Designed meta-learning frameworks using Gaussian process-based synthetic task generation to learn transferable optimization bias and fine-tune surrogates under data scarcity.
- Built probabilistic surrogates for adaptive parameter search in metal additive manufacturing, enabling rapid, defect-free printing of NASA’s GRCop-42 alloy.
- Developed multi-objective, multi-fidelity Bayesian optimization frameworks to jointly optimize energy efficiency, and sensing accuracy in healthcare wearables.
- Designed constrained multi-objective Bayesian optimization for PFAS material discovery, optimizing uptake and selectivity under synthesis feasibility constraints.
- Developing generative optimization methods for offline rare-positive scientific discovery using probabilistic success guidance and data-density modeling.

**Research Intern**, University of Louisville, USA      March 2022 – May 2022

- Explored applications of quantum computing in machine learning and conducted comparative analysis of quantum computing results with traditional approaches.

**Quantum Research Intern**, QWorld (Online)      June 2021 – August 2021

- Implemented hybrid quantum-classical models for image classification using Quantum Convolutional Neural Networks, Ensemble Classification, and Quantum Transfer Learning.

## TEACHING EXPERIENCE

**Graduate Teaching Assistant**, Washington State University, EECS  
Present

January 2023 –

Courses:

- CptS 434/534: Neural Network Design and Applications (Spring 2023, 2025)
- CptS 451: Introduction to Database Systems (Spring 2025)
- CptS 350: Design and Analysis of Algorithms (Fall 2024)
- CptS 440/540: Artificial Intelligence (Spring 2024)
- CptS 570: Machine Learning (Fall 2023)

Responsibilities:

- Developed and delivered tutorials and hands-on labs on neural networks, SGD, fine-tuning, and CNNs/ResNets.
- Provided detailed feedback with clear rubrics, weekly office hours, and targeted help sessions to strengthen student understanding.

**Section Leader**, Code in Place, Stanford University (Online)

April – May 2025

- Taught Python programming to a global cohort through live weekly sessions, emphasizing interactive learning and coding fundamentals.

**Teaching Assistant (Volunteer)**, NVIDIA Deep Learning Institute

April 2025

- Supported certified course labs on neural networks and PyTorch at Washington State University.

## TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, SQL, Bash
- **Machine Learning & Deep Learning:** PyTorch, GPyTorch, BoTorch, scikit-learn, AutoGluon, Transformers (LLMs), Diffusion Models, Gaussian Processes, Bayesian Optimization, Surrogate Modeling, Meta-Learning
- **Data Science & Analytics:** NumPy, pandas, SciPy, Matplotlib, Seaborn; statistical modeling, experiment design, data visualization, feature engineering
- **Systems & Tools:** Linux, Git/GitHub, Conda, Slurm, VS Code, Jupyter, LaTeX/Overleaf
- **Languages:** Arabic (native), French (bilingual), English (professional)

## PROFESSIONAL SERVICE

**Workshop Organizer**

- Co-Organizer, 5th AAAI Workshop on AI to Accelerate Science and Engineering (AI4Science), AAAI 2026. Collaborated with Aryan Deshwal, Jana Doppa, Vipin Kumar, and Carla Gomes.

**Tutorial Organizer**

- Co-instructor, Black-box Optimization from Offline Datasets: Foundations to Recent Advances, Half-day AAAI Tutorial (with Jana Doppa and Aryan Deshwal), AAAI 2026.

### Conference Program Committee

- AAAI (2024, 2026), ICML (2025, 2026), AISI (2026), IAAI (2026, 2027), ICLR (2026), AISTATS (2026), NeurIPS (2026)

### Workshop Reviewer

- AI to Accelerate Science and Engineering (AI4Science) Workshop, AAAI (2026)
- Decision-Making from Offline Datasets to Online Adaptation (DEMO) Workshop, ICML (2026)
- Women in Machine Learning (WiML) Workshop, NeurIPS (2026)

### Journal Reviewer

- IEEE Transactions on Sustainable Computing (2025)

### Leadership & Mentoring

- Graduate Representative, GradSWE, Washington State University (Fall 2025 – Present): Organize the Women in Engineering Speaker Series to inspire and support women in STEM.
- Mentor, AI4ALL Remote Program (May – December 2025): Guiding undergraduate teams through hands-on AI projects; mentored 4 groups to date.
- QWorld (Summer 2021): Assisted participants with programming tasks and answering technical questions.

### Volunteer Service

- Conference Volunteer: ICML (2024), WiML (2024), AAAI (2026)
- Judge, EECS Capstone Poster Competition, Washington State University (Fall 2025)
- Judge, CrimsonCode Hackathon, Washington State University (Spring 2026)

## INVITED TALKS

### Discovery of Feasible 3D Printing Configurations for Metal Alloys via AI-Driven Adaptive Experimental Design

AAAI Conference on Artificial Intelligence and Innovative Applications of AI Conference (AAAI/IAAI) 2026

### Black-box Optimization from Offline Datasets: Foundations to Recent Advances

Quarter-day AAAI Tutorial (with Jana Doppa, Nghia Hoang and Aryan Deshwal) 2026

## PROFESSIONAL DEVELOPMENT

- **Advanced Technical Interview Prep**, CodePath, Spring 2026. Topics: data structures, algorithms, technical interview problem solving, and coding interview practice.
- **Participant – CRA-WP Mentoring Mentors Workshop Series 2025**, *Computing Research Association (Jun–Jul 2025)*:  
5 Multi-session series on evidence-based mentoring practices; topics included inclusive mentorship, aligning expectations, and supporting mentee growth.

- **IEEE Women in Engineering International Leadership Summit (WIE ILS)**, *Tri-Cities, WA*:  
Participated in workshops and panel discussions on leadership and career development, while networking with industry professionals.
- **Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ)**, *tunis, Tunisia*:  
Participated in the program "Promoting Female Talent" Improved soft skills, expanded the professional network, and explored the real-world business environment.